IGCSE 0607 Practice Questions — E1.13 — Combined Questions & Markscheme

Using a calculator

Total Questions: 21 — **Total Marks:** 110 Marks

Table of Contents & Quick Navigation

- **E1.13** Using a calculator
  → **Paper 4 (Calculator)** *(21 Qs)*

1 Calculate the value of

$$\frac{\sqrt{15.2 - 3.4^2}}{4.1 \times 0.25}$$

Write down your full calculator display.

..... **[2]**

2 Work out the value of

$$\frac{\sqrt[3]{25.6 - 3.2^2}}{1.8 \times 0.45}$$

(a) Write down all the figures on your calculator display.

..... **[2]**

(b) Write your answer to part (a) correct to 3 decimal places.

..... **[1]**

3 Consider the following calculation.

(a) Work out the value of

$$\frac{\sqrt{4.1^2 + 5.2}}{2.3^3 - 1.4}$$

Write down all the figures on your calculator display.

..... **[2]**

(b) Give your answer to part (a) correct to 3 significant figures.

..... **[1]**

4 **(a)** Work out the value of

$$\frac{\sqrt{15.2^2 - 8.5^2}}{3.2 \times 1.4}$$

Write down your full calculator display.

..... **[2]**

(b) Write your answer to part (a) correct to 3 significant figures.

..... **[1]**

(c) Calculate

$$\frac{4.5 \times 10^5}{1.2 \times 10^{-3}}$$

Give your answer in standard form.

..... **[1]**

5 **(a)** Consider the following calculation.

Work out the value of

$$\frac{8.3^2 - 4.1}{\sqrt{5.6 + 2.1^3}}$$

Write down all the figures on your calculator display.

..... **[2]**

(b) Give your answer to part (a) correct to 3 significant figures.

..... **[1]**

(c) A computer program runs for 6.45 hours.
Write this time in hours and minutes.

hours ..... , minutes ..... **[1]**

6 **(a)** Work out the value of

$$\frac{4.5^2 - \sqrt{18.2}}{3.1 + 0.8^3}$$

..... **[2]**

(b) Calculate the value of

$$2\sin(48^\circ) - \cos(35^\circ)$$

Give your answer correct to 3 decimal places.

..... **[2]**

7 **(a)** A shopkeeper calculates the cost of an item using the expression

$$\frac{17.3 + 8.14}{5.3}$$

Work out the cost.
Give your answer as a money value in dollars, formatted correctly.

$ ..... **[2]**

(b) A movie lasts for 2 hours and 18 minutes.
Write this time as a decimal number of hours.

..... hours **[2]**

8 **(a)** Use your calculator to find the value of

$$3.5^4 + \sqrt[3]{87.6}$$

Give your answer correct to 2 decimal places.

..... **[2]**

(b) Calculate the value of x when

$$x = \frac{2 \times \sin(43°)}{\cos(18°)}$$

Give your answer correct to 3 significant figures.

$x =$ ..... **[2]**

9 A student uses a calculator to solve three different problems.

(a) The answer to a problem about money is displayed on the calculator as 8.4.
Write this amount appropriately in dollars and cents.

..... **[1]**

(b) The answer to a problem about time is displayed on the calculator as 5.15 hours.
Write this time appropriately in hours and minutes.

hours ..... , minutes ..... **[1]**

(c) A car travels 280 km at an average speed of 85 km/h. Calculate the time taken for this journey.
Give your answer in hours and minutes, correct to the nearest minute.

hours ..... , minutes ..... **[3]**

10 **(a)** Calculate the value of

$$\frac{3.5 + 1.2^2}{\sqrt{4.1 - 2.8}}$$

(i) Write down all the figures on your calculator display.

..... **[2]**

(ii) Write your answer to part (a)(i) correct to 3 significant figures.

..... **[1]**

(b) A calculator displays a journey time as 5.35 hours.
Convert this time to hours and minutes.

h ..... , min ..... **[2]**

11 **(a)** A circular swimming pool has a radius of 4.2 metres.
Calculate the area of the pool.
Give your answer correct to the nearest square metre.

..... m^2 **[2]**

(b) A train journey takes 4 hours and 23 minutes.
The average speed of the train is 78 km/h.

Calculate the distance travelled by the train.
Give your answer correct to 1 decimal place.

..... km **[4]**

12 A plumber charges a fixed fee of \$45 plus an hourly rate of \$32.50.

(a) The plumber works on a job for 3 hours and 40 minutes.
Calculate the total cost of this job.
Give your answer to the nearest cent.

\$ ..... **[3]**

(b) On another job, the total cost charged by the plumber is \$230.25.
Calculate the time the plumber worked, giving your answer in hours and minutes.

hours, minutes **[3]**

13 A metal sphere has a radius of 4.85 cm.

(a) Calculate the volume of the sphere.

The volume, V, of a sphere with radius r is $V = \frac{4}{3}\pi r^3$.

Give your answer correct to 1 decimal place.

..... cm^3 **[2]**

(b) The metal sphere is melted down and made into identical small cones.

Each cone has a radius of 1.2 cm and a height of 3.5 cm.

The volume, V, of a cone with radius r and height h is $V = \frac{1}{3}\pi r^2 h$.

Calculate the maximum number of full cones that can be made.

..... **[4]**

(c) Calculate the percentage of metal wasted from the original sphere.

..... % **[1]**

14 Amir works part-time at a factory.

(a) In one week, Amir works for 24 hours and 40 minutes.
He is paid \$18.60 per hour.
Calculate his total pay for the week.
Give your answer in the correct format for money.

\$ ..... **[3]**

(b) The following month, Amir's total pay increases by 4.5%.
Amir uses his calculator to work out his new total pay.
The calculator displays '479.446'.
Write down Amir's new total pay correct to the nearest cent.

\$ ..... **[1]**

(c) One of the factory machines produces a widget every 3.4 minutes.
Calculate the total time it takes the machine to produce 150 widgets.
Give your answer in hours and minutes.

hours ..... , minutes ..... **[3]**

15 A water tank is in the shape of a sphere with radius 1.4 m.
The tank is completely full of water.

(a) Work out the volume of water in the tank.
Write down all the figures on your calculator display.

..... m^3 **[2]**

(b) Water leaks from the tank at a constant rate of 0.035 cubic metres per minute.
Calculate the time taken for the tank to become completely empty.
Do not round any intermediate values during your working.
Give your answer in hours and minutes, correct to the nearest minute.

hours ..... , minutes ..... **[4]**

16 **(a)** Calculate the exact value of

$$\left(3.2 \times 10^{5}\right) \times \left(4.5 \times 10^{-2}\right)$$

Give your answer in standard form.

..... **[2]**

(b) Work out the value of

$$\frac{1}{\sqrt{2\pi}} \times e^{-0.5^2}$$

Give your answer correct to 4 decimal places.

..... **[2]**

(c) A formula is given by $T = 2\pi\sqrt{\frac{l}{g}}$.

Find the value of l when $T = 1.5$ and $g = 9.81$.

Give your answer correct to 2 decimal places.

$l =$ ..... **[2]**

17 **(a)** Calculate

$$\frac{3.4 \times 10^5 + 2.7 \times 10^4}{8.1 \times 10^{-3}}$$

Give your answer in standard form.

..... **[3]**

(b) A solid metal cone has a base radius of r cm and a height of h cm.
$r = 5.2$ correct to 1 decimal place.
$h = 12.4$ correct to 1 decimal place.
Calculate the upper bound for the volume of the cone.
The volume, V, of a cone with radius r and height h is $V = \frac{1}{3}\pi r^2 h$.

..... cm^3 **[3]**

18 The formula for the period, T seconds, of a simple pendulum is given by

$$T = 2\pi\sqrt{\frac{l}{g}}$$

where l is the length of the pendulum in metres and g is the acceleration due to gravity, 9.81 m/s^2.

(a) A pendulum has a length of 85 cm.
Calculate the period T.
Give your answer correct to 3 decimal places.

$T =$ ..... seconds **[2]**

(b) Rearrange the formula to make l the subject.

$l =$ ..... **[2]**

(c) A different pendulum has a period of exactly 2.5 seconds.
Calculate the length of this pendulum in centimetres.
Give your answer correct to 1 decimal place.

..... cm **[3]**

19 The volume, V, of a spherical cap is given by the formula

$$V = \frac{1}{3}\pi h^2(3R - h)$$

where R is the radius of the sphere and h is the height of the cap.

(a) Calculate V when $R = 8.45$ cm and $h = 3.2$ cm. Give your answer correct to 1 decimal place.

$V =$ ..... cm^3 **[2]**

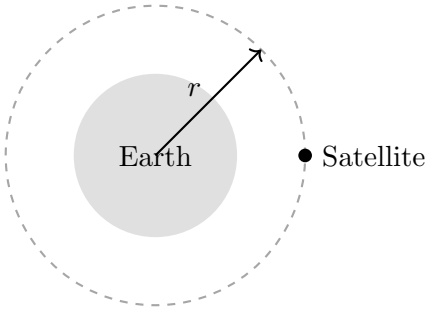
(b) Rearrange the formula to make R the subject.

$R =$ ..... **[3]**

(c) A different spherical cap has volume $V = 500$ cm^3 and height $h = 5.6$ cm. Calculate the radius, R, of this sphere.
Give your answer correct to 3 significant figures.

$R =$ ..... cm **[2]**

20 A satellite is in a circular orbit around the Earth.



(a) The speed, v m/s, of the satellite is given by the formula

$$v = \sqrt{\frac{GM}{r}}$$

where:

- $G = 6.67 \times 10^{-11}$
- $M = 5.97 \times 10^{24}$
- $r = 6.80 \times 10^{6}$

Work out the value of v.
Give your answer correct to 3 significant figures.

$v =$ ..... **[3]**

(b) The time, T seconds, taken for the satellite to complete one full orbit is given by

$$T = \frac{2\pi r}{v}$$

Using the exact value of v from your calculator, calculate the time taken for one full orbit.
Give your answer in hours, minutes and seconds, correct to the nearest second.

hours ..... , minutes ..... , seconds ..... **[4]**

21 The volume, V, of a spherical cap is given by the formula

$$V = \frac{1}{3}\pi h^2(3R - h)$$

(a) Calculate the volume V when $R = 8.74$ and $h = 2.31$.
Give your answer correct to 1 decimal place.

$V =$ ..... **[3]**

(b) A student calculates V by first rounding the values of R and h to 1 decimal place before substituting them into the formula.
Calculate the difference between this incorrect answer and the exact volume.
Give your difference correct to 2 significant figures.

Difference = ... **[4]**

1 Numerator $\approx 1.907878...$, Denominator $= 1.025$ **[M1]**

1.861344784 **[A1]**

[Back to Question] *[Total Marks: 2]*

2 (a) 3.068822... **[B2]**

If 0 scored, allow [B1] for seeing 15.36 or 2.485... or 0.81

(b) 3.069 [B1] (strict FT from their (a) if given to at least 4 d.p.)

[Back to Question] *[Total Marks: 3]*

3 (a) Numerator $\sqrt{22.01} = 4.69148\ldots$ or Denominator $12.167 - 1.4 = 10.767$ seen **[M1]**

0.435727839 **[A1]**

(b) 0.436 **[B1]**

[Back to Question] *[Total Marks: 3]*

4 (a) M1 for $\sqrt{158.79}$ or $12.60119\ldots$ or 4.48 seen

2.812765825 (accept to at least 4 decimal places) **[A1]**

(b) 2.81 **[B1ft]**

(c) 3.75×10^8 **[B1]**

[Back to Question] *[Total Marks: 4]*

5 (a) 64.79 or 14.861 or ≈ 3.855 seen **[M1]**

16.80675747 (allow up to 2 missing trailing digits depending on calculator model) **[A1]**

(b) 16.8 **[B1]**

(c) 6 hours 27 minutes **[B1]**

[Back to Question] *[Total Marks: 4]*

6 (a) 4.423... or $\dfrac{15.98...}{3.612}$ **[M1]**

4.42 or 4.424 **[A1]**

(b) 1.4862... − 0.8191... or 0.6671... **[M1]**

0.667 **[A1]**

[Back to Question] *[Total Marks: 4]*

7 (a) 4.8 **[M1]**

4.80 **[A1]**

(b) $2 + \frac{18}{60}$ or $\frac{138}{60}$ **[M1]**

2.3 **[A1]**

[Back to Question] *[Total Marks: 4]*

8 **(a)** $150.0625 + 4.4412...$ or $154.5037...$ **[M1]**

154.50 **[A1]**

(b) $\frac{1.3639...}{0.9510...}$ or $1.434...$ **[M1]**

1.43 **[A1]**

[Back to Question] *[Total Marks: 4]*

9 **(a)** 8.40 **[B1]**

(b) 5 hours 9 minutes **[B1]**

(c) $280 \div 85$ **[M1]**

3.294... **[A1]**

3 hours 18 minutes **[A1]**

[Back to Question] *[Total Marks: 5]*

10 **(a)** **(i)** Numerator 4.94 or denominator 1.14... seen **[M1]**

4.332666... **[A1]**

(ii) 4.33 **[B1]**

(b) 0.35×60 **[M1]**

5 h 21 min **[A1]**

[Back to Question] *[Total Marks: 5]*

11 **(a)** $\pi \times 4.2^2$ or $55.417...$ **[M1]**

55 **[A1]**

(b) $23 \div 60$ or $0.3833...$ seen **[M1]**

Time $= 4 + \frac{23}{60}$ or $4.3833...$ hours **[M1]**

Distance $= 78 \times 4.3833...$ **[M1]**

341.9 **[A1]**

[Back to Question] *[Total Marks: 6]*

12 **(a)** Time $= 3 + \frac{40}{60}$ or $\frac{11}{3}$ hours **[M1]**

Cost $= 45 + 32.50 \times \frac{11}{3}$ **[M1]**

164.17 **[A1]**

(b) $230.25 - 45 = 185.25$ **[M1]**

$\frac{185.25}{32.50} = 5.7$ hours **[M1]**

5 hours 42 minutes **[A1]**

[Back to Question] *[Total Marks: 6]*

13 **(a)** M1 for $\frac{4}{3} \times \pi \times 4.85^3$

477.8 **[A1]**

(b) M1 for $\frac{1}{3} \times \pi \times 1.2^2 \times 3.5$ (implies $5.2778\ldots$)

M1 for $\frac{\text{their } 477.8}{\text{their } 5.2778\ldots}$

A1 for $90.5\ldots$

90 **[B1]**

(c) 0.584 (accept 0.582 to 0.585) **[B1ft]**

[Back to Question] *[Total Marks: 7]*

14 **(a)** Time $= 24\frac{40}{60}$ or 24.666... or $\frac{74}{3}$ **[M1]**

$\frac{74}{3} \times 18.60 = 458.8$ **[M1]**

458.80 (must have trailing zero) **[A1]**

(b) 479.45 **[B1]**

(c) 3.4×150 or 510 minutes **[M1]**

$510 \div 60$ or 8.5 **[M1]**

8 hours 30 minutes **[A1]**

[Back to Question] *[Total Marks: 7]*

15 **(a)** $\frac{4}{3}\pi(1.4)^3$ **[M1]**

11.49404... **[A1]**

(b) their 11.494... $\div 0.035$ **[M1]**

328.4... or 328 (minutes) **[A1]**

their 328.4... $\div 60$ (converting to hours) **[M1]**

5 hours 28 minutes **[A1]**

[Back to Question] *[Total Marks: 6]*

16 **(a)** 14.4×10^3 or 14400 **[M1]**

1.44×10^4 **[A1]**

(b) $\frac{1}{2.5066...} \times 0.7788...$ or $0.398942... \times 0.778800...$ or $0.310696...$ **[M1]**

0.3107 **[A1]**

(c) $1.5 = 2\pi\sqrt{\frac{l}{9.81}} \implies \frac{1.5}{2\pi} = \sqrt{\frac{l}{9.81}} \implies 0.2387... = \sqrt{\frac{l}{9.81}}$ **[M1]**

$l = 9.81 \times (0.2387...)^2 \implies l = 0.55909...$ **[M1]**

0.56 **[A1]**

[Back to Question] *[Total Marks: 6]*

17 **(a)** Numerator $= 340000 + 27000 = 367000$ or 3.67×10^5 **[M1]**

$3.67 \times 10^5 \div (8.1 \times 10^{-3}) = 45308641.98...$ **[M1]**

4.53×10^7 or 4.531×10^7 **[A1]**

(b) Upper bound for $r = 5.25$ and Upper bound for $h = 12.45$ **[M1]**

$V = \frac{1}{3} \times \pi \times 5.25^2 \times 12.45$ **[M1]**

359.34... or 359 **[A1]**

[Back to Question] *[Total Marks: 6]*

18 **(a)** M1 for $2\pi\sqrt{\frac{0.85}{9.81}}$

1.850 **[A1]**

(b) M1 for $\left(\frac{T}{2\pi}\right)^2 = \frac{l}{g}$ or $\frac{T^2}{4\pi^2} = \frac{l}{g}$

$\frac{gT^2}{4\pi^2}$ **[A1]**

(c) M1 for their $l = \frac{9.81 \times 2.5^2}{4\pi^2}$

A1 for $l = 1.5532\ldots$ metres

155.3 **[A1]**

[Back to Question] *[Total Marks: 7]*

19 **(a)** $\frac{1}{3} \times \pi \times 3.2^2 \times (3 \times 8.45 - 3.2)$ **[M1]**

237.5 **[A1]**

(b) $\dfrac{3V}{\pi h^2} = 3R - h$ **[M1]**

$\dfrac{3V}{\pi h^2} + h = 3R$ **[M1]**

$R = \dfrac{V}{\pi h^2} + \dfrac{h}{3}$ or $R = \dfrac{3V + \pi h^3}{3\pi h^2}$ **[A1]**

(c) $\dfrac{500}{\pi \times 5.6^2} + \dfrac{5.6}{3}$ **[M1]**

6.94 **[A1]**

[Back to Question] *[Total Marks: 7]*

20 **(a)** $\dfrac{6.67 \times 10^{-11} \times 5.97 \times 10^{24}}{6.80 \times 10^6}$ or $5.8558... \times 10^7$ **[M1]**

$\sqrt{5.8558... \times 10^7}$ or 7652.36... **[M1]**

7650 **[A1]**

(b) Total distance $= 2 \times \pi \times 6.80 \times 10^6$ or 42725660... **[M1]**

Time in seconds $= \dfrac{42725660...}{7652.36...} = 5583.32...$ **[M1]**

$5583 \div 3600 = 1.55...$ (implies 1 hour) and remainder 1983 seconds **[M1]**

1 hour 33 minutes 3 seconds **[A1]**

[Back to Question] *[Total Marks: 7]*

21 **(a)** $\dfrac{1}{3} \times \pi \times 2.31^2 \times (3 \times 8.74 - 2.31)$ **[M1]**

133.607... **[A1]**

133.6 **[A1]**

(b) Rounding initial values: $R = 8.7$ and $h = 2.3$ **[B1]**

$\dfrac{1}{3} \times \pi \times 2.3^2 \times (3 \times 8.7 - 2.3)$ or 131.84... **[M1]**

Difference $= 133.607... - 131.844... = 1.76...$ **[M1]**

1.8 **[A1]**

[Back to Question] *[Total Marks: 7]*